

MOSES YEBEI

A.I AUTOMATION & SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software engineer with 7+ years shipping and operating production systems, specialising in applied AI. Builds the full path from architecture to production: distributed services in Python and TypeScript, Temporal workflow orchestration, Kafka and ClickHouse pipelines, and Kubernetes on GKE with Terraform and Helm. On the AI side, has designed multi-agent architectures, knowledge-graph retrieval layers, RAG pipelines, and fine-tuned models, with evaluation and cost instrumentation treated as part of the system rather than an afterthought. Holds engineering standards through code review and schema contracts, and works directly with non-technical stakeholders — recent work cut invoice processing from ~15 days to under an hour.

SKILLS

Technical Skills

- **AI & LLM Systems:** AI Agents, Agentic Workflows (LangGraph, CrewAI, LangChain), RAG Pipelines, Embedding-Based Retrieval, Semantic Search, Fine-Tuning, Document Classification, Computer Vision, Prompt Versioning
- **Automation & Integration:** MCP, Enterprise Connectors, ERP Integration, Microsoft 365/SharePoint, Workflow Orchestration (Temporal), RPA (PowerApps), API & Middleware Integration, Approval Workflows, Audit Logging
- **Machine Learning:** Scikit-learn, PyTorch, XGBoost, LightGBM, Time-Series Forecasting, Anomaly Detection, Feature Engineering, Model Evaluation
- **MLOps & Observability:** MLflow, Weights & Biases, Langfuse, OpenTelemetry, Grafana, CI/CD, Automated Testing
- **Data Engineering:** Polars, PySpark, Kafka, ClickHouse, ETL Pipelines, Fabric
- **Databases:** PostgreSQL, FalkorDB/Cypher, Redis, Vector DBs (Chroma, FAISS, Pinecone, Weaviate), SQL & NoSQL
- **Cloud & Infra:** GCP (GKE, Vertex AI), AWS (EC2, S3, Lambda), Azure ML Studio, Terraform, Helm, Docker, Kubernetes
- **Languages:** Python, TypeScript, JavaScript, SQL

PROFESSIONAL EXPERIENCE

Global Business Solutions, TechnoBrain

Oct 2025 – Present | Nairobi/Remote

AI & Automation Project Lead - Finance & Audit · 2026 – Present

Own the AI and automation roadmap for finance and audit, working with finance, audit, and procurement stakeholders to identify and prioritise opportunities.

Finance Automation: Built an invoice pipeline (ERP export → AI field mapping → validation → report generation → email distribution) cutting processing from ~15 days to under 1 hour. Extended the pattern to management reporting and payslip distribution.

Audit & Compliance: Automated auditor reporting, including carbon audit and Microsoft reporting, with review checkpoints and an audit trail on every document.

Tender Intelligence: Monitored Kenyan Government and UN procurement portals with email summaries, prequalification matching, and submission tracking; automated proposal, SLA, and payment request generation.

Reusable AI Skills: Replaced one-off prompting with AI Skills encapsulating business logic, templates, mapping rules, and validation, backed by a knowledge base of SOPs and business rules. Set approval workflows, prompt versioning, and audit logging as standards.

Roadmap: Driving MCP as the standard integration layer, connectors to ERP/SharePoint/Microsoft 365/CRM, enterprise RAG with vector search, and agentic workflows in LangGraph. Packaging internal automations into commercial suites for government, NGO, and enterprise clients.

Tech Stack: Python, Gemini/Gems, Claude, ChatGPT, Codex, MCP, LangGraph, Enterprise RAG, ERP integration, Microsoft 365/SharePoint, Vapi

Autar.ai Project - Founding AI/Data Engineer (Seattle, remote)

Founding engineer on the applied-AI platform behind an automated software analysis product, working with CEO Jared Green from architecture to production.

AI Agent Systems: Built the data layer serving 7 parallel specialist agents tailored knowledge-graph slices instead of raw context, cutting token usage and improving output quality. Tracked per-agent cost, latency, and quality via Langfuse and PostHog.

Knowledge Graph Architecture: Designed a three-layer pipeline (FalkorDB/Cypher knowledge base with 149 entity types, per-repo state extraction, LLM-enriched analysis) enabling structured querying across large, messy codebases.

Data Pipelines: Engineered 20+ Temporal workflows and 120+ activities (7 concurrent extractors, 20+ languages) with Kafka streaming and a ClickHouse warehouse feeding PostgreSQL, FalkorDB, and vector stores.

Engineering Standards: Reviewed merge requests across the team, using code review to spread platform conventions — schema contracts, workflow patterns, observability.

Reliability: Shipped safely despite non-deterministic LLM extraction by enforcing schema validation at write time, then hardening with quality dashboards. Owned observability (Grafana, OpenTelemetry) and production Kubernetes on GKE (Terraform, Helm, Vault).

Tech Stack: Python, TypeScript, PyTorch, Temporal, FalkorDB, PostgreSQL, ClickHouse, Kafka, LangChain, LiteLLM, Langfuse, GKE, Terraform, Grafana, OpenTelemetry, Docker

AI Engineer & Software Architect, AICE Africa

Jan 2025 – Sep 2025 | Nairobi

Owned a user-facing, full-stack AI application end to end, from requirements through design, implementation, testing, deployment, and maintenance with active users. The platform automated the job application lifecycle using coordinated AI agents.

Full-Stack Delivery: Designed and built the whole stack: a FastAPI backend with custom middleware, a multi-agent system coordinated by a central orchestrator, and data services on Supabase (PostgreSQL) and Redis. Containerized with Docker, established a Pytest test suite, and deployed and maintained the running service.

Applied ML & NLP: Built the intelligence layer using LangChain and the Gemini API for the LLM workflows, a FAISS-backed RAG pipeline for semantic search, and scikit-learn, spaCy, and NLTK for CV parsing, entity extraction, and intent detection that routed user requests to the right agent.

Reliability: Added a circuit-breaker pattern for graceful degradation and designed the codebase around clear separation of concerns and dependency injection, keeping it maintainable as agents were added.

Tech Stack: Python, FastAPI, LangChain, Google Gemini, FAISS, scikit-learn, spaCy, NLTK, Supabase (PostgreSQL), Redis, Docker, Pytest

AI & Forensic Engineer, Litigation Data Solutions, SALIX Data

Aug 2022 – Dec 2024

Technical lead on forensic and e-discovery delivery for high-stakes litigation, working directly with CEO Rob Adams to align the AI roadmap with client and business needs and translating investigative findings into decisions for legal clients and leadership.

Text Analytics at Scale: Developed an NLP and ML pipeline processing 1M+ documents with 99% accuracy for classification, extraction, and search, and a speech-to-text pipeline transcribing legal audio at 99% accuracy to speed client turnaround.

Forensic Data Pipelines: Architected and maintained pipelines across Relativity, CloudNine Law, and Microsoft Purview for data mining, extraction, and documentation, improving process efficiency by 30% and cutting retrieval time by 25%.

Domain-Expert Collaboration: Gathered requirements directly from litigators and delivered tooling and reporting that turned technical output into decisions non-technical users could act on.

Tech Stack: Python, NLP, Machine Learning, Text Analytics, Speech-to-Text, Relativity, CloudNine Law, Microsoft Purview, Data Pipeline Development

Data Science & Analytics Consultant, Upwork and Fiverr

Jan 2021 – Nov 2023

- Provided data science and analytics consulting services to diverse organizations across fintech, banking, agriculture, supply chain, e-commerce, healthcare, and microfinance sectors.
- Developed and deployed predictive models and data-driven solutions, improving operational efficiency by 40% for clients.
- Designed and implemented a fraud detection model for banks and fintech companies, reducing fraudulent transactions by 75% and minimizing financial losses.
- Engineered and deployed AI-powered chatbots for real estate agents, improving lead engagement and response time by 60%, leading to higher conversion rates.
- Conducted in-depth customer interaction analytics across online platforms (customer reviews, ad clicks, social media, website interactions), optimizing marketing strategies and boosting customer retention by 35%.
- Delivered user training sessions and workshops on analytics dashboards, tools, and data interpretation, empowering clients and teams to leverage data-driven decision-making.
- Tracked customer engagement across trending topics on social media, analyzing interactions, sentiment, and ad performance. Delivered real-time dashboards that provided actionable insights, leading to a 20% improvement in targeted marketing campaigns and a 15% increase in customer retention.
- **Technical Skills:** Python, Machine Learning, Predictive Modeling, Data Visualization, SQL, Chatbot Development, Customer Analytics, Anomaly Detection

Data-Driven Budget Analyst, Bomet County

Mar 2018 – Jan 2020

- Managed financial reporting and reconciliation for projects exceeding 5 billion KSH, ensuring compliance and accuracy in public funding.
- Developed and implemented data-driven forecasting models that informed county budgeting and strategic planning, improving resource allocation efficiency by 15%.
- Played a key role in financial advisory teams, providing critical support for county projects and initiatives through data analysis and insights.
- **Technical Skills:** Financial Reporting, Budgeting, Forecasting, Financial Analysis, Data Analysis, Statistical Modeling



ARTIFICIAL INTELLIGENCE PROJECTS

LLM Fine-Tuning for Code & Generative AI

2024 – 2024

Fine-tuned a 1.5B-parameter generative model, improving relevance and coherence in text and Python code generation. *PyTorch, open-source LLMs, Python.*

Reinforcement Learning for Trading Strategies

2024 – present

RL agent optimising investment strategies, outperforming rule-based baselines in simulated environments. *Python, TensorFlow, Keras, OpenAI Gym, Financial APIs.*

Automatic Speech Recognition, Speech 2 Text GenAI

2024 – 2024

Automated YouTube lectures and videos into text for data analysis and note-taking using OpenAI speech recognition models. Enhancing accessibility and usability of video content for educational and analytical purposes.

Computer Vision , Real-Time Object Detection and Speed Estimation

2023 – 2024

Vehicle detection and speed estimation for smart-city traffic management, deployed against live highway footage. *YOLOv8, OpenCV, Python.*

Additional Data/automation projects

- **Vector Database for RAG Pipelines:** Built a vector database from scratch for RAG pipelines, enabling efficient retrieval of information for question answering and document summarization.
- **Data Engineering for UAE Ministry of Investment:** Utilized PySpark to move data from UN Comtrade for the UAE Ministry of Investment, facilitating data-driven decision-making.
- **RPA for UAE Khalifa Fund:** Implemented RPA solutions using PowerApps for the UAE Khalifa Fund, automating processes and improving efficiency.



EDUCATION / TECHNICAL COURSES

BSc. Actuarial Science, Karatina University

2012 – 2016

Application of mathematical, statistical, probabilistic, and financial theories to solve future business problems.

DP-100 Azure Machine Learning, Udemy

Sep 2024 – Dec 2024

Azure ML and AutoML for data integrity, training, and deploying scalable ML models.

Data Science & AWS, ALX Africa

2023 – 2024

In-depth technical skills in SQL, data analysis, web development, and machine learning across numerous real-world projects.

Tools: Google Sheets, Excel, SQL, Python, Regression Analysis, Machine Learning, NLP, and AWS.

Machine Learning , Stanford University - Coursera

2021 – 2022

A subfield of AI that leverages data and intersects with statistics, cognitive science, information theory, and probability theory.

Deep Learning, Duke University - Coursera

2021 – 2022

Completed auditing a deep learning course at Duke University. Focusing on Deep Q Learning and its underlying mathematics.

The course emphasized the significance of deep learning in advancing AI capabilities, particularly in decision-making and problem-solving applications.

Google Cloud Fundamentals, Google Qwiklabs

2022 – 2022

Gained in-depth understanding of Google Cloud products, services, tools, features, benefits, and use cases, including cloud concepts and capabilities.



REFERENCES

Bonaventure Tonui, *Solutions Architect and Cloud Developer*, Amazon - AWS

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